C-SASO: A Clustering-Based Size-Adaptive Safer Oversampling Technique for Imbalanced SAR Ship Classification

Yongxu Li, Xudong Lai, Mingwei Wang, and Xi Zhang

***Abstract*— Ship classification using synthetic aperture radar (SAR) induces effective marine applications but suffers from imbalanced datasets. Common ship types (majority classes) contain many more instances than rare ship types (minority classes), resulting in performance loss since minority instances tend to be ignored. Meanwhile, it is difficult and time-consuming to produce reliable instances for rare ships. The synthetic minority oversampling techniques have shown great potential to balance the distribution of classes by synthesizing minority instances. However, the newly synthesized instances may overlap with the majority instances, reducing the separability among classes, or being far away from the classification boundary, which is meaningless. This article proposes a clustering-based size-adaptive safer oversampling (C-SASO) technique to address the imbalanced classification problem. Proven schemes, including cluster minority instances, adaptively allocate oversampling sizes, and assign weights are adopted for selecting minority instances. Then, new instances are synthesized according to the safe metric of selected instances, instead of randomly inserted. Furthermore, multiple handcrafted features are tested to provide a clearer classification boundary. The proposed comprehensively considers the usefulness and safety of synthesizing instances. Experiments on the OpenSARShip and the FUSAR-Ship datasets demonstrate that the proposed technique achieves significantly better results.**

***Index Terms*— Handcrafted features, imbalanced dataset, oversampling, ship classification, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active observation system that has the advantage of operating day and night in different weather conditions. SAR can quickly capture large-scale high-resolution images and is widely researched in maritime [1]. Ship detection and classification from SAR images have been a topic of interest in recent decades [2]–[16]. Ship classification provides comprehensive information for more efficient marine applications, such as marine surveillance and security. Many high-quality images are available with advanced spaceborne SAR systems, but low-efficiency interpretations have become a bottleneck and a limiting factor. Accurate and robust ship classification is practically needed and has received increasing attention.

Previous studies from the literature were limited to a few verified ships without enough public datasets to compare their performance [6]–[16]. In 2017, the medium-resolution OpenSARShip dataset was built based on the Sentinel-1 constellation [17]. Thereafter, the high-resolution FUSAR-Ship dataset based on the Gaofen-3 satellite was created and published in 2020 [18]. These two datasets facilitate the ship classification research [19]–[29], but they are highly imbalanced. In OpenSARShip, bulk carriers and container ships have 811 and 212 chips in the dataset, respectively. These two classes are the main components of merchant ships, both subcategories of cargo. Meanwhile, cargos occupy approximately 75% of the entire dataset, while some classes have only single digits. Similarly, the number of categories and subcategories is imbalanced in the FUSAR-Ship. Imbalanced data pose a significant challenge for classification because minority classes become hard to learn, and more minority instances are necessary [27]–[30].

The production of numerous reliable instances for rare ship types is difficult and complex, consuming a lot of labor and time [31]. The synthetic minority oversampling technique (SMOTE) is proposed to balance the distribution between classes by synthetizing minority instances; it randomly selects a minority instance and then interpolates new instances between that instance and its N nearest minority instances [32]. However, the newly generated instances will aggravate the overlap between classes since the technique does not consider majority instances distribution. Various approaches have been proposed to address this problem, which can be summarized as improving the instance selection scheme and improving the synthetic instance generation scheme. The former applies different measures to find valuable instances that are easily misclassified. The Borderline SMOTE determines the borderline between classes and only applies oversampling to the minority instances distributed on the borderline to address this shortcoming [33]. It may fail to identify the borderline in some scenarios in that the minority instances are sparsely distributed. The Cluster SMOTE divides the minority instances into subclusters and then implements SMOTE within every single subcluster to avoid overlapping with majority instances [34]. The minority instances are

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divided by the unsupervised K-means clustering algorithm, and there may be majority instances in the final minority subclusters. The adaptive semi-unsupervised weighted oversampling (A-SUWO) technique approaches this problem by presenting a semi-unsupervised clustering algorithm; it checks whether the two minority subclusters to be merged contain any majority instances [35]. In addition to the imbalance that exists among classes (between-class imbalance), performance degrades when there is an imbalance among subclusters, some of which have fewer instances than others. Since hard-to-learn instances that contain essential information usually belong to small subclusters that are easily ignored and thus require more oversampling [36]. In the A-SUWO technique, the sampling number is further adaptively allocated depending on the classification complexity of each subcluster.

The abovementioned oversampling techniques improve the instance selection scheme rather than the synthetic instance generation scheme. New instances are arbitrarily inserted between elaborately selected instances. When new minority instances are generated at the boundaries of classes (or subclusters), some new instances may be too close to the majority instances. The Safe-Level SMOTE counts the distribution of instances in each minority instance's neighbor as a safe level (sl) metric to synthesize new instances reasonably [37]. New instances are positioned closer to the ones with a bigger sl, so that all new instances are generated in a safer region. But there is an over-fitting caused by new instances that are more likely far from the classification boundary; meanwhile, the within-class imbalance is not considered [38]. Since the sl only guides the final interpolation process, it can be directly integrated into some of the above techniques to complement each other.

Under imbalanced scenarios, another strategy is optimizing feature space [39], [40]. A well-established feature set provides a more apparent class boundary, improving the classification performance. It also promotes the implementation of the oversampling technique since these techniques are implemented in the given feature space. Previous works from the literature on ship classification have developed various features to extract the interactions between incident radar waves and the structure of different ships [6]–[16]. Geometric features such as length and width, echo scattering features such as local radar cross section (LRCS) and scattering center, texture features such as scale-invariant feature transform (SIFT) descriptors and the SAR-SIFT descriptors have been widely studied and proved efficient with different SAR images. Besides, as representative discrimination features, the Old Lincoln features describe the difference between targets in size, shape, intensity, and spatial distribution [41]; the New Lincoln features capture the changes in the spatial dispersion of high-intensity pixels [42]; Bhanu's features catch shape-related divergence [43]; Gao's features measure the contrast between high- and low-intensity pixels [44]; their potential for ship classification is worth to be further studied.

In this article, a clustering-based size-adaptive safer oversampling (C-SASO) technique is proposed to address the imbalance distribution of SAR ship classification. The synthetic minority oversampling technique is employed to balance the datasets and proven instance selection schemes are adopted, including clusters of the minority instances, adaptively allocating the size of subclusters, and assigning weight to each minority instance. Then, new instances are reasonably interpolated according to the sl of selected instances. Furthermore, a well-established feature set is determined by evaluating the classification ability of multiple representative features to improve the interclass separation. Comparison experiments prove the efficiency of the proposed technique, and the results are recorded by a confusion matrix to provide more detail.

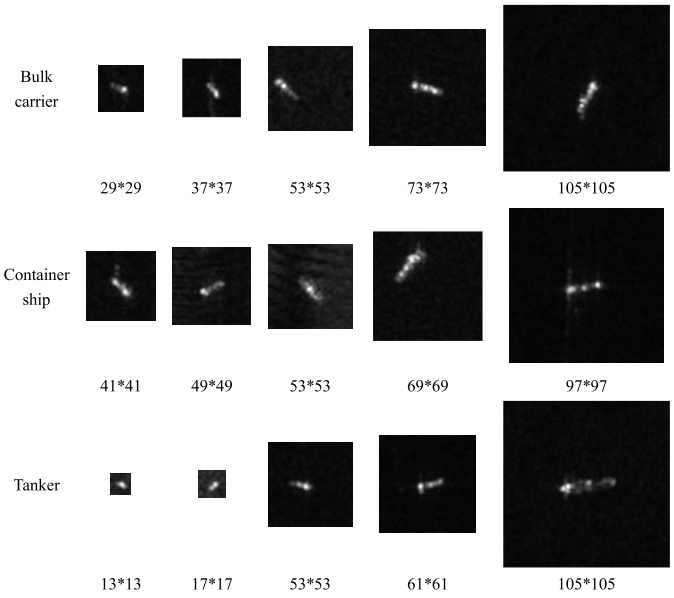


Fig. 1. Ship chips of various sizes from the OpenSARShip.

The remainder of the article is organized as follows. Section II describes the two imbalanced datasets and the features extracted for classification. Section III presents a detailed introduction of the proposed oversampling technique. Section IV shows the experimental design and the comparative analysis of the results. Finally, Section V provides the conclusion.

II. Imbalanced Datasets and Features for Ship Classification

A. Datasets

1) OpenSARShip: The OpenSARShip was constructed from Sentinel-1 SAR images. They are level-1 single look complex and level-1 ground range detected (GRD) products in interferometric wide swath mode with dual polarization (VV and VH). The GRD product has the approximately same pixel spacing and resolution in both range and azimuth direction that can retain the geometric features of the ship targets, many studies have been conducted with the GRD product [22]–[26]. However, the size of ships is quite different, with some ships only occupying a few pixels at the medium-resolution of 20×22 m^2, lacking adequate description, as shown in Fig. 1. Moreover, the size of chips significantly varies, from the smallest 13×13 pixels to the largest 185×185 pixels, causing difficulty in applying some classification approaches. Besides, the dataset is highly imbalanced where some classes have thousands of chips, while

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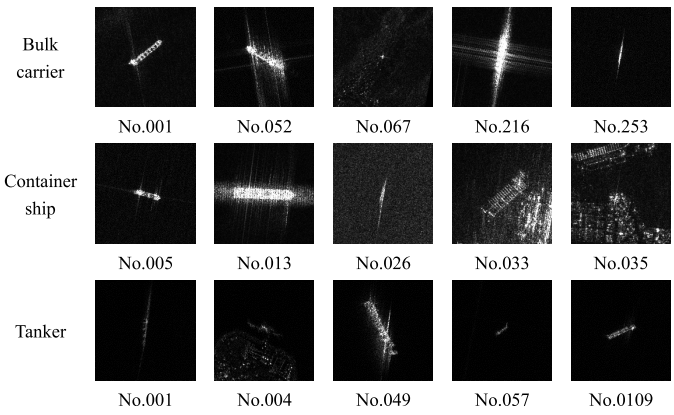


Fig. 2. Ship chips with complex background from the FUSAR-Ship.

others have only single digits, which is a significant challenge for classification.

Consistent with earlier research [6]–[13] and also convenient the comparison with some studies based on this dataset [17], [25], [26], three main merchant ship types are selected: bulk carrier, container ship, and tanker. These ships cover around 80% of the international shipping market [21], [22]. There are 811 bulk carriers, 212 container ships, and 173 tankers, with an imbalanced ratio of 4.69:1.22:1 [45]. The number of instances of three classes is slightly different from the article of OpenSARShip. This is caused by the classified ship types being a bit coarse, and two chips of the same target under VV and VH polarization are taken as one instance to extract more characteristics.

2) FUSAR-Ship: The FUSAR-Ship dataset was extracted from 126 images acquired by the Gaofen-3 satellite in ultrafine strip-map mode. It has a higher resolution, with an azimuth resolution of 1.124 m and a range resolution ranging from 1.7 to 1.754 m. It provides a fixed size of each chip as 512×512 pixels, and the higher resolution presents more detailed information, as shown in Fig. 2. However, the size of ships also widely varies, and there are nonship interferential objects within a small number of chips. The minimum enclosing rectangle information of each target was not provided. It is hard to extract ships from these chips even manually, such as 067 of bulk carriers, 035 of container ships, and 004 of tankers. Besides, there is also an imbalance between different ship types. Similarly, three ship types, 273 bulk carriers, 148 tankers, and 65 container ships are collected for classification experiments, with an imbalanced ratio of 4.2:2.28:1 [46].

B. Commonly Used SAR Image Features

Previous works from the literature on ship classification have developed some valuable features, including geometric features, Hu moment invariants features, scattering center features, principal axis features, LRCS features, LRCS (three sections) features, SIFT features, and SAR-SIFT features have been tested on the OpenSARShip dataset, where the LRCS (three sections) features and SAR-SIFT features achieved better results [17], [23]. However, the published overall accuracy (OA) provides limited information for the objective evaluation of this serious imbalanced dataset. In addition, Old Lincoln, New Lincoln, Gao's, and Bhanu's features describe targets in size, shape, scatter energy, spatial distribution, and texture, which have successfully distinguished ships from other nonship objects, and their potential for ship classification has not been fully verified.

The standard deviation feature, fractal-dimension feature, weighted-rank fill ratio feature, and specific entropy feature of the Old Lincoln features are calculated within a minimum enclosing rectangle of targets. In some cases, such as strong ship targets with large sidelobes, the minimum enclosing rectangle contains extraneous background pixels that may bias the precise calculation of these features [10], [27]. In the proposed study, these features are directly extracted within a target-shaped blob which is obtained by the constant false alarm rate (CFAR) method [47], while the step of calculating the minimum enclosing rectangle is omitted. The new feature set is denoted as Old-Lincoln-B by replacing these four features from the Old Lincoln features. The LRCS (three sections) and SAR-SIFT features are selected along with Old Lincoln, Old-Lincoln-B, New Lincoln, Gao's, and Bhanu's features to produce the experiments, where the extraction of SAR-SIFT features employs the bag-of-words model [48]. The adopted features are summarized in Table I.

III. C-SASO Technique

A. Overall Scheme

The minority oversampling technique is adopted to balance the datasets, and the C-SASO technique is proposed to improve the effectiveness of newly generated instances and ensure safety. The proposed C-SASO technique includes three main steps as shown in Fig. 3: 1) semi-unsupervised clustering as Fig. 3(b); 2) adaptive subcluster sizing as Fig. 3(c); and 3) synthetic instances generation in safer regions as Fig. 3(d) and (e). The minority classes of the dataset are first divided into subclusters and then the sampling number is adaptively allocated for each subcluster, finally, different weights are assigned to each instance within one subcluster. These proven robust instance selection schemes are entirely inherited. Furthermore, to improve the synthetic instance generation scheme, new instances are located in safer regions that avoid exacerbating the overlapping among classes. The basic principles and specific operations are detailed as described in Sections III-B–III-D.

B. Semi-Unsupervised Clustering

The agglomerative complete-linkage hierarchical clustering (ACHC) algorithm is widely used, it generates clusters in a bottom-up fashion that does not require the number of clusters to be prefixed [49]. For ACHC, each instance is set as a subcluster initially, and a threshold T is set as the average of the median Euclidean distance among all instances of the same class. Thereafter, the two closest subclusters are found, and if the distance between them is less than T, they are merged into one subcluster. Then, the distances of all existing subclusters are recalculated and repeat the above operation until the distance between all subclusters is greater than T. The threshold T is calculated as follows:

$$T = d_{\text{avg}} * C_{\text{thres}} \tag{1}$$

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TABLE I
ADOPTED SAR IMAGE FEATURES FOR CLASSIFICATION

Name	Summary	Dimension
LRCS (3 sections)	The sum, the mean, and the standard of the values extracted from the bow, middle, and stern sections of ship	9
SAR-SIFT	A SIFT-like descriptors specifically dedicated to SAR imaging	128
Old-Lincoln	Standard deviation, Fractal dimension, Weighted-rank fill ratio, Mass, Diameter, Square-normalized rotational inertia, Maximum CFAR statistic, Mean CFAR statistic, Percent bright CFAR statistic, Specific-entropy, Contiguousness	16
Old-Lincoln-B	Standard deviation-B, Fractal dimension-B, Weighted-rank fill ratio-B, Mass, Diameter, Square-normalized rotational inertia, Maximum CFAR statistic, Mean CFAR statistic, Percent bright CFAR statistic, Specific-entropy-B, Contiguousness	16
New-Lincoln	Threshold, Activation, Dispersion, Inflection, Acceleration	5
Gao	Average signal-to-noise-ratio, Peak signal-to-noise-ratio, Percentage of bright Pixels	3
Bhanu	Projection, Distance, Moment	10

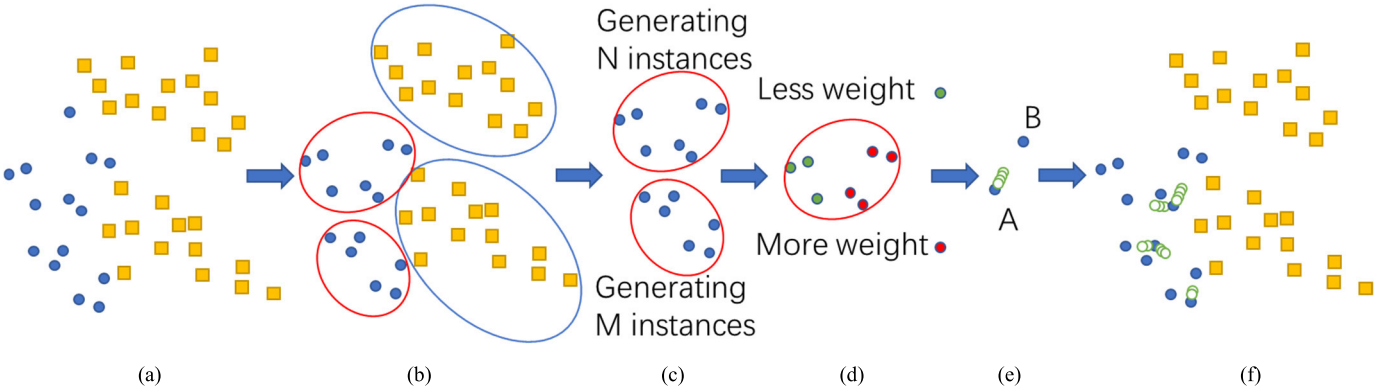


Fig. 3. Processes of C-SASO technique. (a) Imbalanced dataset. (b) Semi-unsupervised clustering. (c) Allocating subcluster sizing, $M = N$ in usual, here it is adaptively determined. (d) Assigning weights to instances within one subcluster, red points get more weight than green points since they are closer to classification boundaries. (e) Synthetic instances generation in safer regions, suppose that point A is safer than point B, and then new instances will be interpolated closer to point A. (f) Final balanced dataset.

where d_{avg} is the average of $d_{\text{med},h}$, and $d_{\text{med},h}$ represents the median Euclidean distance between all instances of the same class. C_{thres} is the coefficient to tune the threshold for clustering.

When clustering the minority class, majority instances may be included in the final minority subclusters, since it is unsupervised and the distribution of the majority instances is not considered, as shown in Fig. 4. The semi-unsupervised ACHC algorithm checks whether the two minority subclusters to be merged contain any majority instances. First, it records the classes of the N_n nearest neighbors for each instance. If the classes of these neighbors are all different from the instance, the instance is taken as noise and then removed. Thereafter, the traditional ACHC algorithm is used to cluster majority instances and get m majority subclusters $C\max j_{i=1,\ldots,m}$. For minority instances, before two subclusters are merged, an additional step is to check whether there is a majority subcluster between them. If so, they will never be merged. These procedures are described in Algorithm 1.

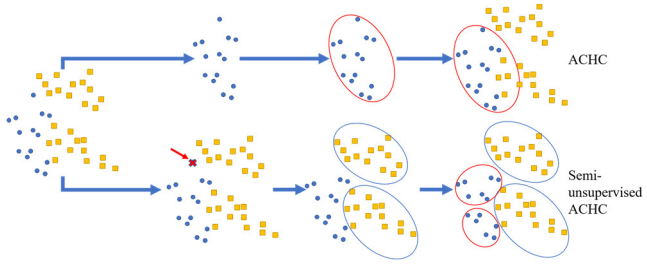


Fig. 4. Clustering minority instances, ACHC versus the semi-unsupervised ACHC.

C. Adaptive Subcluster Sizing

Some subclusters may easily be misclassified that need more oversampling, while others do not. The classification complexity is calculated to balance minority subclusters sampled size dynamically. The classification error rate is used as a measure and calculated with K-fold cross-validation on each subcluster. As an example, take any two subclusters $C\min_a$, $C\min_b$, the final size of each subcluster is proportional to their

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Algorithm 1 Semi-Unsupervised ACHC

Inputs:

-I: Original dataset to be oversampled.

-c_{thres}: Coefficient to tune the threshold T for clustering.

-N_n: Number of nearest neighbors used to identify noisy instances.

Outputs:

-O: Minority subclusters $Cmin_{j=1,\ldots,n}$.

Procedure:

1. Remove noisy instances from the dataset.
2. Determine T using (1).
3. Cluster majority class, which will result in m subclusters $Cmaj_{i=1,\ldots,m}$.
4. Assign each minority instance to a separate sub-clutter.
5. Find the two closest subclusters $Cmin_a$ and $Cmin_b$.
6. Check if there is any overlapping majority subcluster between $Cmin_a$ and $Cmin_b$.
7. If yes, set their distance to infinity and return to step 5. Else, merge $Cmin_a$ and $Cmin_b$ into one subcluster $Cmin_c$.
8. Recalculating the Euclidean distance between $Cmin_c$ and existing subclusters.
9. Repeat steps 5 to 8 until the Euclidean distance between the closest subclusters is no less than T.

Algorithm 2 Adaptive Subcluster Sizing

Inputs:

-I: Minority subclusters

-K: Number of folds in the K-fold cross-validation

Outputs:

-O: Final sizes S for all minority subclusters $Cmin_{j=1,\ldots,n}$

Procedure:

1. Randomly split each minority subcluster into K folds.
2. Build a classifier using $K-1$ folds of each minority subcluster in addition to all majority instances as the training set.
3. Test the classifier using the remained fold of each minority subcluster.
4. Repeat steps 2 to 3 K times.
5. Determine standardized average minority error rate $\bar{\varepsilon}_a^*$ using (3).
6. Determine final sizes S for all subclusters using (2).

standardized average error rate. The procedure is described in Algorithm 2

$$\frac{S_a}{S_b} = \frac{\bar{\varepsilon}_a^*}{\bar{\varepsilon}_b^*} \quad \forall a,\ b \in \{1, \ldots, n\} \tag{2}$$

where S_a and S_b are the final size of $C\min_a$, $C\min_b$, and $\bar{\varepsilon}_a^*$, $\bar{\varepsilon}_b^*$ are the standardized average error rate of $C\min_a$, $C\min_b$. $\bar{\varepsilon}_a^*$ is obtained using the following equation:

$$\bar{\varepsilon}_a^* = \frac{\varepsilon_a^*}{\sum_{a=1}^{n} \varepsilon_a^*} \tag{3}$$

where ε_a^* is the average error rate of $C\min_a$. Let ε_{ak} represents the number of minority samples is mistakenly divided into the majority in fold K of $C\min_a$, and error rate ε_{ak}^* is acquired by dividing ε_{ak} by the size of $C\min_a$, and then ε_a^* is obtained by averaging the error rate of all folds.

Algorithm 3 Synthetic Instances Generation in Safer Regions

Inputs:

-I: Minority subclusters

-N_w: Number of nearest majority neighbors to be found for minority instance to determine the weights.

-N_i: Number of nearest minority neighbors to be found for selected minority instance to generate synthetic instance.

-N_s: Number of nearest neighbors to be found for selected minority instance to calculatesl.

-S: Final sizes S for all subclusters

Outputs:

-O: Oversampled dataset.

Procedure:

I. Determine the probability distribution for instances within each minority subcluster:

-For each subcluster $Cmin_{j=1,\ldots,n}$

1. For all minority instances in subcluster $Cmin_j$, find N_w nearest majority neighbors.
2. Determine w for each minority instance in $Cmin_j$ using (5) and by estimating TH_j.
3. Transform the weights to a probability distribution P_j using (6).

II. Oversample minority instances with a sl

4. Select a minority instance a in subcluster $Cmin_j$ for sampling according to P_j.
5. Randomly select one of its N_s nearest neighbors b that belongs to the same subcluster.
6. Computer sl_a and sl_b
7. if $sl_b \neq 0$
   $sl_ratio = sl_a/sl_b$
   else $sl_ratio = \infty$
   calculate β with (9).
8. $c = a + \beta * (a - b)$
9. Add c to dataset O.
10. Repeat steps 4 to 9 until the subcluster size reaches S_j.

D. Synthetic Instances Generation in Safer Regions

1) Assigning Weights: The generation of synthetic instances is completed in two steps according to the final size of each minority subcluster. The first step is to individually assign weights to all instances in one subcluster according to their distance to the nearest N_w majority instances. More weights are set for the instances closer to the majority instances, which are more likely to be misclassified. For example, h in the subcluster $C\min_i$ ($C\min_i$ with R instances), the distances between it and the N_w nearest majority instances are $d(v)$ ($v = 1, \ldots, N_w$). The standardized $d'$is obtained by dividing d by the features' dimension D. Considering that a larger d' means the instance is far away from the majority of instances that should be given a small weight. Similarly, a smaller d' should be given a relatively large weight. Therefore, the reciprocal of d' is more suitable as a guide. In addition,

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TABLE II
RESULTS (%) FOR OVERSAMPLING TECHNIQUES APPLIED TO DIFFERENT FEATURES OF OPENSARSHIP

Features	Measures	Original	SMOTE	Borderline SMOTE	Safe-Level SMOTE	Cluster SMOTE	Cluster Safer SMOTE	A-SUWO	C-SASO
LRCS (3 sections)	*OA*	78.93	**84.42**	80.60	83.19	77.09	77.17	80.75	83.16
	F	76.65	**84.95**	81.46	83.09	76.55	77.89	80.19	81.19
SAR-SIFT	*OA*	84.20	84.28	84.45	87.46	87.71	87.63	87.65	**88.25**
	F	83.32	84.75	85.10	87.57	87.94	87.76	87.53	**88.00**
New Lincoln	*OA*	82.11	82.36	82.53	83.79	78.18	80.27	83.19	**85.12**
	F	80.48	82.71	83.10	83.07	77.59	79.66	82.50	**85.14**
Gao	*OA*	76.42	64.97	63.29	76.52	69.40	70.48	75.68	**77.93**
	F	74.36	65.59	64.32	76.19	69.54	69.71	75.37	**77.88**
Bhanu	*OA*	79.93	77.51	75.75	80.65	74.00	75.67	78.61	**81.52**
	F	76.89	78.48	76.65	80.77	73.97	75.95	78.90	**81.77**
Old Lincoln	*OA*	85.03	80.77	83.70	**85.87**	81.94	83.36	84.54	84.12
	F	84.16	81.64	84.38	**85.88**	81.61	82.65	84.42	83.77
Old Lincoln-B	*OA*	85.87	87.54	87.79	85.95	85.70	87.96	86.87	**89.05**
	F	85.16	87.85	88.15	86.08	85.67	87.64	86.68	**88.76**

to prevent the reciprocal of d' from being abnormally large when the distance between two instances is very close, the definition of the closeness factor between two instances is as follows:

$$\Gamma(v) = f_i(d') \tag{4}$$

where

$$f_i(x) = \begin{cases} x, & \text{if } x \le \mathrm{TH}_i \\ \mathrm{TH}_i, & \text{otherwise} \end{cases}$$

as the cutoff function, when the reciprocal of d' is greater than the threshold TH_i, it will be set as the threshold TH_i. The threshold TH_i is set as the average of the reciprocal of d' between all instances in the same subcluster to their most adjacent majority instance. TH_i is then recalculated in different subclusters.

The weight w of instance h is the sum of N_w closeness factors as (5). The probability distribution P_i is obtained by dividing each weight by the sum of all instances' weights in a subcluster as (6). Then, two instances, a and b are selected to oversample, a is selected according to the P_i, and b is randomly selected from the N_i nearest instances of a that belong to the same subcluster. Thereafter, the synthetic instance c is usually randomly interpolated between a and b as (7)

$$w(h) = \sum_{v=1}^{N_w} \Gamma(v) \tag{5}$$

$$P_i(h) = \frac{w(h)}{\sum_{h=1}^{R} w(h)} \tag{6}$$

$$c = a + \beta * (a - b) \tag{7}$$

where β is a random number between 0 and 1.

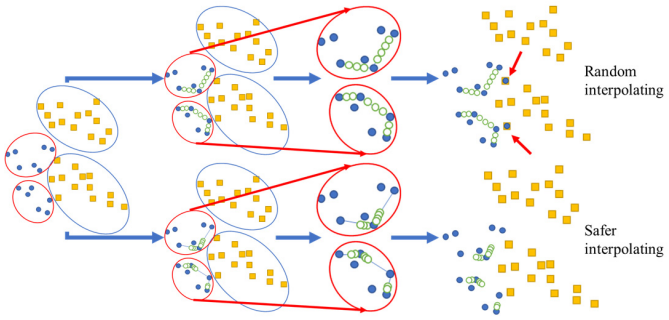


Fig. 5. Synthesizing new minority instances, random interpolating versus safer interpolating. (the hollow green circles represent new minority instances, and the yellow rectangles with a blue ball inside represent majority instances that may be misclassified).

2) Calculating sl and Synthesizing Instances: Fig. 5 shows the insertion of new instances between two selected minority instances, and the random insertion affects the correct judgment of the majority instances. To further lighten these influences, as shown in the second procedure of Algorithm 3, the C-SASO calculates the sl and sl ratio (sl_ratio) with (8) and then generates instances closer to a bigger sl, that is, only in safer regions. The sl is the number of minority instances in N_s nearest neighbors

$$\mathrm{sl_ratio} = \mathrm{sl}_a / \mathrm{sl}_b. \tag{8}$$

Specifically, for different sl_ratio, five corresponding cases and different β settings guide the generation. The first case of $\mathrm{sl_ratio} = \infty$ and $\mathrm{sl}_a = 0$ means that both a and b are noises, depicting that a synthetic instance will not be generated; the second case of $\mathrm{sl_ratio} = \infty$ and $\mathrm{sl}_a \neq 0$ means b is noise and the synthetic instance will be generated by duplicating a that far away from noise b; the third case of $\mathrm{sl_ratio} = 1$ means that a is as safe as b and synthetic instances

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TABLE III
RESULTS (%) FOR OVERSAMPLING TECHNIQUES APPLIED TO DIFFERENT FEATURES OF FUSAR-SHIP

Features	Measures	Original	SMOTE	Borderline SMOTE	Safe-Level SMOTE	Cluster SMOTE	Cluster Safer SMOTE	A-SUWO	C-SASO
LRCS (3 sections)	*OA*	74.28	63.37	74.38	79.01	73.25	72.43	79.22	**79.84**
	F	72.46	63.83	75.47	78.24	71.71	72.78	77.90	**79.00**
SAR-SIFT	*OA*	72.58	80.86	75.51	75.93	75.72	74.28	75.10	**81.89**
	F	70.74	81.31	75.34	76.44	75.70	74.23	75.35	**82.31**
New Lincoln	*OA*	82.30	76.75	77.57	**86.01**	79.22	81.48	85.19	85.39
	F	79.60	76.75	78.14	**86.00**	79.02	81.63	84.66	84.81
Gao	*OA*	77.78	72.43	65.74	75.10	66.67	71.40	74.90	**77.98**
	F	75.09	73.96	66.19	75.96	67.40	68.23	74.84	**78.71**
Bhanu	*OA*	76.75	73.46	71.19	77.16	67.49	69.55	74.07	**77.98**
	F	74.44	73.77	71.94	77.02	67.49	68.84	74.11	**78.30**
Old Lincoln	*OA*	82.92	87.45	**87.65**	86.63	81.07	82.30	86.21	86.01
	F	82.10	87.90	**87.98**	86.17	81.72	82.56	86.27	85.57
Old Lincoln-B	*OA*	85.19	87.24	87.24	86.42	86.21	87.04	87.45	**88.07**
	F	84.67	87.82	87.97	86.60	86.42	87.15	87.41	**88.09**

TABLE IV
ORIGINAL RESULTS (%) OF DIFFERENT FEATURES ON OPENSARSHIP

LRCS (3 sections)				Bhanu				Gao			
	B	C	T		B	C	T		B	C	T
B	92.85	04.69	02.47	B	**95.93**	02.34	01.73	B	92.11	05.18	02.71
C	66.51	27.83	05.66	C	72.64	24.53	02.83	C	63.68	33.96	02.36
T	20.81	02.89	76.30	T	27.17	00.00	72.83	T	44.51	00.58	54.91
OA	78.93			*OA*	79.93			*OA*	76.42		
F	76.65			*F*	76.89			*F*	74.36		
Old Lincoln				New Lincoln				SAR-SIFT			
	B	C	T		B	C	T		B	C	T
B	95.31	03.58	01.11	B	94.45	03.33	02.22	B	94.08	04.32	01.60
C	45.28	51.89	02.83	C	57.55	38.68	03.77	C	46.70	49.53	03.77
T	19.08	03.47	**77.46**	T	21.39	01.16	**77.46**	T	15.03	04.62	80.35
OA	85.03			*OA*	82.11			*OA*	84.20		
F	84.16			*F*	80.48			*F*	83.32		
Old Lincoln-B											
	B	C	T								
B	95.31	03.08	01.60								
C	41.04	**56.60**	02.36								
T	20.81	01.73	**77.46**								
OA	**85.87**										
F	**85.16**										

Note: B for bulk carrier, C for container ship, and T for tanker. The bold values in the table denote the optimal result.

will be randomly interpolated between a and b; the fourth case of sl_ratio $>$ 1 means that a is safer than b and synthetic instances will be generated closer to a by setting parameter β in the range [0, 1/sl_ratio]; the fifth case of sl_ratio $<$ 1 means that b is safer than a and synthetic instances will be generated closer to b by setting parameter β in the range [1 − sl_ratio, 1].

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TABLE V
ORIGINAL RESULTS (%) OF DIFFERENT FEATURES ON FUSAR-SHIP

LRCS (3 sections)				Bhanu				Gao			
	B	C	T		B	C	T		B	C	T
B	84.98	07.33	07.69	B	90.11	01.36	01.97	B	91.21	06.59	02.20
C	78.46	16.92	04.62	C	72.31	05.66	02.83	C	83.08	09.23	07.69
T	18.24	02.03	79.73	T	20.95	01.16	66.47	T	14.86	02.03	83.11
OA	74.28			OA	76.75			OA	77.78		
F	72.46			F	74.44			F	75.09		
Old Lincoln				New Lincoln				SAR-SIFT			
	B	C	T		B	C	T		B	C	T
B	90.84	02.47	00.62	B	**93.77**	02.20	04.03	B	89.01	03.66	07.33
C	64.62	10.38	00.47	C	72.31	16.92	10.77	C	64.62	24.62	10.77
T	06.76	02.89	76.88	T	06.08	04.05	89.86	T	31.08	05.41	62.84
OA	82.92			OA	82.30			OA	72.58		
F	82.10			F	79.60			F	70.74		
Old Lincoln-B											
	B	C	T								
B	90.48	07.33	02.20								
C	49.23	**44.62**	06.15								
T	04.73	02.03	**93.24**								
OA	**85.19**										
F	**84.67**										

Note: B for bulk carrier, C for container ship, and T for tanker. The bold values in the table denote the optimal result.

The generated synthetic instance is forced to be closer to the instance with a higher sl

$$\beta = \begin{cases} \emptyset, & \text{the 1st case} \\ 0, & \text{the 2nd case} \\ \text{random}(0, 1), & \text{the 3rd case} \\ \text{random}\left(0, \dfrac{1}{\text{sl}_{\text{ratio}}}\right), & \text{the 4th case} \\ \text{random}(1 - \text{sl}_{\text{ratio}}, 1), & \text{the 5th case.} \end{cases} \tag{9}$$

IV. EXPERIMENTAL STUDY

The experiments are implemented using MATLAB R2021a on a 64-bit Windows 10 system with an AMD Ryzen 7-2700 CPU and 8 GB RAM. Multiple sets of comparative experiments are implemented to evaluate the proposed oversampling technique, all tests are performed with fivefold cross-validation, and the confusion matrix is used to record the result.

The proposed C-SASO is compared with SMOTE, Borderline SMOTE, Safe-Level SMOTE, Cluster SMOTE, and A-SUWO. The sl is also integrated with Cluster SMOTE to guide the generation of synthetic instances, referred to as Cluster Safer SMOTE. These oversampling techniques are implemented for each extracted feature set, obtaining multiple test datasets. The classification results of these data provide valuable information for comprehensively verifying the capacity of these oversampling techniques and determining a well-established feature set. Since the synthetic minority instances are more likely to be correctly recognized, only the classification results of the original instances are counted.

The parameters related to the number of nearest neighbors like N_n, N_w, N_i, and N_s are set to 5 for all oversampling techniques. c_{thres} is set to 1 and the K is set to 3 for A-SUWO and C-SASO. These parameter settings referred to the recommended value [20]. The F-measure (F) and OA are used to evaluate the results. The F-measure is calculated using the following equations, with γ is set to 1:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

$$F_{\text{measure}} = \frac{\left(1 + \gamma^2\right) * P * R}{\gamma^2 * R + P} \tag{12}$$

where TP denotes the number of true positives, FN denotes that of false negatives, and FP denotes that of false positives.

A. Overall Results of the Two Datasets

The comparative experiments are carried out on two imbalanced SAR ship classification datasets. The classification results are summarized in Tables II and III, respectively. The optimal performance of each feature set is highlighted in bold.

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TABLE VI
RESULTS (%) OF OPENSARSHIP DATASET BALANCED BY DIFFERENT OVERSAMPLING TECHNIQUES

SMOTE				Borderline SMOTE				Safe-Level SMOTE			
	B	C	T		B	C	T		B	C	T
B	86.44	10.23	03.33	B	85.45	10.48	04.07	B	89.27	07.89	02.84
C	15.09	84.43	00.47	C	08.96	**89.62**	01.42	C	23.11	74.53	02.36
T	03.47	00.00	**96.53**	T	03.47	00.00	**96.53**	T	12.14	03.47	84.39
OA	87.54			OA	87.79			OA	85.95		
F	87.85			F	88.15			F	86.08		
Cluster SMOTE				Cluster Safer SMOTE				A-SUWO			
	B	C	T		B	C	T		B	C	T
B	90.63	05.92	03.45	B	94.70	03.58	01.73	B	92.36	05.06	02.59
C	26.42	68.40	05.19	C	32.55	66.98	00.47	C	30.66	66.98	02.36
T	10.40	05.78	83.82	T	16.18	01.73	82.08	T	10.98	03.47	85.55
OA	85.70			OA	87.96			OA	86.87		
F	85.67			F	87.64			F	86.68		
C-SASO				Old Lincoln-B							
	B	C	T		B	C	T				
B	**94.82**	03.33	01.85	B	95.31	03.08	01.60				
C	30.19	68.40	01.42	C	41.04	56.60	02.36				
T	12.14	00.58	87.28	T	20.81	01.73	77.46				
OA	**89.05**			OA	85.87						
F	**88.76**			F	85.16						

Note: B for bulk carrier, C for container ship, and T for tanker. The bold values in the table denote the optimal result.

The results of these two datasets show that the oversampling techniques played a positive role in classification; the optimal performance is achieved after implementing oversampling. It is observed that some oversampling techniques have large deviations in the effects of different feature sets. They play a positive role in some feature sets, while others are the opposite. For example, some oversampling techniques are implemented on the LRCS (three sections), New Lincoln, Gao, Bhanu, and Old Lincoln features, and the classification performance has deteriorated. While in the SAR-SIFT and Old Lincoln-B features, all oversampling algorithms improved classification accuracy. Safe-Level SMOTE and C-SASO techniques perform robustly, and C-SASO is superior to other oversampling techniques on most features. Cluster Safer SMOTE and C-SASO perform better in most cases than Cluster SMOTE and A-SUWO, respectively, since they adopt sl to guide the generation of new instances.

Despite the wide variation in ship size and chip size, and some chips accompanying noise, all features yielded good results. The OA of the LRCS (three sections) and SAR-SIFT features (78.93%, 84.20%) is improved compared to the published results from Huang's work (75.69%, 80.12%) [11], because more characteristics are provided by dual-polarization. New Lincoln, Old Lincoln, SAR-SIFT, and Old Lincoln-B features achieved high accuracy on the OpenSARShip dataset. New Lincoln, Old Lincoln, and Old Lincoln-B features performed well on the FUSAR-Ship dataset, but SAR-SIFT is poor. Among them, the OA of Old-Lincoln features is 85.03% for OpenSARShip and 82.92% for FUSAR-ship and that of Old Lincoln-B features is 85.87% and 85.19%. There are 0.84% and 2.27% higher than Old-Lincoln features since four features are revised to focus more on pixels of the target itself. The F-measure of these datasets also shows that Old Lincoln-B features (85.16% and 84.67%) are 1.00% and 2.57% higher than Old Lincoln features (84.16% and 82.10%), respectively. It can be inferred that more minority instances are correctly classified.

For these two extremely imbalanced datasets, further analyses are necessary with the help of the confusion matrix to study the performance of different features and different oversampling techniques.

B. Comparison Between Different Features

The classification results of all extracted features before oversampling are collected to evaluate their performance. The OA provides limited information. Hence, confusion matrices recorded detailed results to comparatively analyze different features, as shown in Tables IV and V. The optimal performance is highlighted in bold.

The classification performance of each type of ship can be observed with the confusion matrix. There are common phenomena in the results of both datasets. The bulk carrier contains more instances that obtain higher accuracy, and the

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TABLE VII
RESULTS (%) OF FUSAR-SHIP DATASET BALANCED BY DIFFERENT OVERSAMPLING TECHNIQUES

SMOTE				Borderline SMOTE				Safe-Level SMOTE			
	B	C	T		B	C	T		B	C	T
B	84.98	13.55	01.47	B	83.88	15.02	01.10	B	88.28	08.79	02.93
C	18.46	**80.00**	01.54	C	23.08	76.92	00.00	C	33.85	63.08	03.08
T	04.73	00.68	94.59	T	00.68	01.35	**97.97**	T	02.70	04.05	93.24
OA	87.24			OA	87.24			OA	86.42		
F	87.82			F	87.97			F	86.60		
Cluster SMOTE				Cluster Safer SMOTE				A-SUWO			
	B	C	T		B	C	T		B	C	T
B	87.55	11.36	01.10	B	88.28	10.26	01.47	B	89.38	08.79	01.83
C	38.46	58.46	03.08	C	35.38	60.00	04.62	C	36.92	60.00	03.08
T	02.70	01.35	95.95	T	02.03	01.35	96.62	T	03.38	00.68	95.95
OA	86.21			OA	87.04			OA	87.45		
F	86.42			F	87.15			F	87.41		
C-SASO				Old Lincoln-B							
	B	C	T		B	C	T				
B	**90.11**	08.42	01.47	B	90.48	07.33	02.20				
C	32.31	63.08	04.62	C	49.23	44.62	06.15				
T	03.38	01.35	95.27	T	04.73	02.03	93.24				
OA	**88.07**			OA	85.19						
F	**88.09**			F	84.67						

Note: B for bulk carrier, C for container ship, and T for tanker. The bold values in the table denote the optimal result.

container ship and the tanker have fewer instances in which accuracy is low. It can be speculated that due to the low-class separability and high imbalance of the two datasets, the classifier may ignore the minority class to optimize the OA.

The results of the OpenSARShip dataset indicate that the SAR-SIFT features achieve the optimal accuracy of tankers, 80.35%. The Old Lincoln-B features correctly identified most of the container ships and also obtained the best recognition performance, the accuracy of container ships is improved by 4.71%, compared with the Old Lincoln features. The order of these features according to the accuracy is Old Lincoln-B, Old Lincoln, SAR-SIFT, New Lincoln, Bhanu, LRCS (three sections), and Gao.

Unlike the results of OpenSARShip, most tankers of FUSAR-Ship are correctly identified. The Old Lincoln-B features and New Lincoln features are significantly higher at 93.24% and 89.86%. The results of the FUSAR-Ship dataset also manifest that the Old Lincoln-B features perform better than other features and achieve an optimal effect on container ships. It still maintains a higher recognition accuracy for bulk carriers and that for container ships and tankers is significantly improved by 34.24% and 16.36%, respectively, compared with the Old Lincoln features. The order of these features according to the accuracy is Old Lincoln-B, Old Lincoln, New Lincoln, Bhanu, Gao, LRCS (three sections), and SAR-SIFT. Which consistent with the performance on the OpenSARShip dataset, except that SAR-SIFT performs poorly. In general, the Old Lincoln-B features achieve better classification performance on both datasets, providing more significant interclass differences.

C. Comparison Between Different Oversampling Techniques

Based on the Old Lincoln-B features, the classification results of two datasets after implementing oversampling are recorded by confusion matrices to study different oversampling techniques, as shown in Tables VI and VII. The optimal performance is highlighted in bold.

The accuracy of minority instances improved significantly with the SMOTE and the Borderline SMOTE. In the OpenSARShip dataset, the accuracy of container ships achieves 89.62% and that of tankers is 96.53%. In the FUSAR-Ship dataset, the optimal result for container ships is 80% and that for tankers is 97.97%. However, these two techniques lead to a significant drop in the accuracy of identifying bulk carriers, which is the lowest among all oversampling techniques. Because the generated synthetic instances are heavily overlapping with the majority of instances, more bulk carriers are mistakenly classified as container ships or tankers, meaning more false positives mixed in the latter's results. In contrast, Safe-Level SMOTE creates new instances in safer regions that reduce the impact on the majority class to a certain extent.

The results of the other four oversampling techniques show that the accuracy of the majority class is higher. Simultaneously, the accuracy of the two minority classes significantly improved. The proposed technique effectively maintains the accuracy of the majority class, and it is only 0.49% and 0.37%

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lower than before oversampling on two datasets, respectively. The OA and F-measure of A-SUWO are better than Cluster SMOTE, which proves the helpful schemes including more accurate clustering, considering the imbalance within the classes, and giving more weight to instances closer to the majority class. In addition, based on these two techniques, sl is introduced to guide the final generation of new instances, instead of randomly interpolating. The results of the two techniques that employ sl are better than those without sl. C-SASO achieves superior results with an OA of 89.05% for OpenSARShip, and FUSAR-Ship is 88.07%, which are 3.18% and 2.88% higher than before oversampling. Specifically, the recognition accuracy of container ships is increased by 11.8% and 18.46% and that of tankers is 9.82% and 2.03%. Meanwhile, the F-measure increased from 85.16% and 84.67% to 88.76% and 88.09%, respectively, increasing 3.60% and 3.42%.

V. Conclusion

In this article, a C-SASO technique is proposed to optimize the imbalance problem of SAR ship classification. Proven valuable schemes are adopted to select valid minority instances to synthesize necessary new instances. Then, the sl metric is used to guide the final interpolation process to avoid the new instances overlapping with the majority instances. The classification ability of commonly used and some representative discrimination features are investigated, to increase interclass differences. Four features belonging to the Old Lincoln features are revised and outperform other features. Experiments on OpenSARShip and FUSAR-Ship datasets demonstrate that the proposed oversampling technique effectively balances these datasets. The accuracy of minority classes is improved, while the majority class still maintains its high level. The superior classification performance is achieved by the proposed C-SASO technique oversampling on the Old Lincoln-B features, reaching an OA of 89.05% on OpenSARShip and 88.07% on FUSAR-Ship. The results also prove that the features commonly used to distinguish false alarms are competent in classifying different types of ships, which supports the implementation of false alarm discrimination and ship classification simultaneously. Future work will focus on the integration of these two critical steps of automatic ship target recognition, reducing processes and improving efficiency.

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